Class Structure

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How does this class work?

This course is guided by the philosophy that learning mathematics requires focused effort, reflection, and revision. Assessment of your learning is based on a proficiency system, in which you have multiple chances to demonstrate proficiency, with opportunities to reflect on and improve previous understanding.

0.1 Attendance and Engagement

Since we will be doing much of our learning through collaborative in-class activities, it's important that you to come to class prepared and ready to engage. This means being on-time, completing preview activities prior to class, working productively on in-class activities, asking questions, and listening to and supporting your classmates.

While I do expect you to be in class, I also recognize that life happens and occasionally things come up that might prevent you from being there. Unless it becomes a habit, there is no penalty to missing class apart from missing out on that day's activity. If you do miss class, I expect you to check the class Canvas page to see what we covered that day and/or come see me with your questions.

Finally, if you find yourself missing class more than a few times, you can expect that I will check with you to see how I can support your attendance better.

0.2 Preview and In-Class Activities

Most of the time when we begin a new section from the book (approximately every other class period), I will assign a short **Preview Activity** to be completed prior to class. These are designed to introduce and motivate new material. We will begin class by discussing these activities, so it's important that you come to class having completed it before hand.

Some sections, however, we will be skimming or even skipping entirely. In those cases, I will not assign a preview activity.

Note that assigned preview activities may not be made up. They are designed to support our day-to-day work, and if you miss class they no longer serve that purpose.

Much of our in-class time will be spent working in small groups on additional activities designed to build up our understanding. While I won't generally collect these, I will from time-to-time to help me monitor class progress and student engagement.

0.3 Homework

To assist you in building your understanding with the course material and in practicing new skills, I will assign short weekly homework sets. We will set aside some time in class, to discuss these problems in our groups.

In addition, you have access to an online homework system - called **WebWork** - to give you additional practice as needed. This homework will not be graded for correctness - but completion can be used as evidence of engagement.

Instructions on how to use WebWork will be posted on Canvas.

0.3.1 Weekly Engagement Report

Every week, I will ask you to complete a short reflection assignment summarizing your engagement with the course for that week. Your report should include the following information:

- Did you complete the Preview Activites and the assigned homework problems this week? Did you find them helpful?
- What questions came up for you while doing these problems or doing the class activities? Did you get stuck on any particular problems and if so, how did you resolve these questions?
- What questions do you still have?
- How confident are you about your understanding of this week's materal?

0.4 Labs

Approximately every 2 weeks, I'll assign a lengthier homework set that is designed to give you a chance to engage with the material more deeply. These problems are meant to be challenging and you should expect to have questions! They might also introduce new concepts or ask you to apply what you've learned to new situations.

In addition to giving you an opportunity to extend your engagement with the mathematical content of the course, these assignments are also a chance to practice communicating your mathematical thinking. It's not enough to solve a problem if you can't convince someone else that you know what you're talking about!

0.5 Due Dates and Extensions

Just like in the "real world", due dates in this class exist and are important. They help all of us plan our lives and stay organized. That said, there is usually a certain amount of flexibility and so if something comes up that's going to make it difficult (or impossible) to complete an assignment on time, just let me know. Simply send me an email letting me know when you expect to get the assignment to me.

- If you need to ask for an extension, please let me know in a timely manner (preferably before the due date).
- If you ask for an extension, you should plan on handing in work within one week of the due date.
- You do not need to give me a reason for your extension request. I trust that if you are asking for an extension then you have a reason!
- There is no penalty for late work except that you might not get timely feedback on your work. This is especially relevant for labs which can be revised and resubmitted.
- If you ask for many extensions, you can expect that I will reach out to see if we can work together to find ways for you to keep up with the work in the course.

There are, however, two **hard deadlines** to be aware of: Oct. 9th (Friday before Fall break) and Dec. 4th (last day of classes). Except in unusual circumstances or by prior arrangement, I will not accept work after those dates.

Finally, note that since the purpose of the Preview Activities is to prepare you for that day's class, these may not be handed in late.

0.6 Collaboration

Unless otherwise instructed, I encourage you to work together with classmates on homework or other assignments although any work that you hand in should be your own. You should also acknowledge your collaborators. Unless otherwise instructed, all check-ins should be done individually.

0.7 Weekly Check-Ins

Most weeeks (typically Tuesdays) we will have a short check-in that focuses on 1-3 recently covered topics. This is the primary mechanism for you to demonstrate proficiency on our specific learning targets - every learing target will appear at least 3 in-class check-ins.

For each Learning Target that appears on a check-in, I will provide feedback as follows:

P Proficient. You have demonstrated sufficient understanding with that particular topic.

R Minor Revisions Needed. You have demonstrated an almost complete understanding, but there are minor errors or gaps in your explanation.

N Not Yet Proficient. More significant errors or gaps indicate that you are still working on building a sufficient understanding of this topic.

I Incomplete. Fragmentary or missing response.

0.7.1 Revisions

If a learning target is marked as needing minor revisions (**R**) you may write up a complete, corrected solution and hand it in **within one week**. Your revision should include:

- A brief sentence explaining what your error was.
- A brief description of the steps you took to correct your mistakes.
- A complete, rewritten solution. Do not simply write over your original problem.

Staple your new solution **on top of** your original check-in and indicate clearly that this is a Revision. If you successfully correct your errors, I will convert your **R** to a **P**.

0.7.2 Re-Assessemnts

Each Learning Target will appear on at least three in-class check-ins (including the midterm and final check-ins) so if you aren't able to demonstrate proficiency the first time a Learning Target appears, you will have additional chances in class. In addition, you may also show proficiency outside of class:

- You may do a re-assessment during drop-in hours. Note that during drop-in hours, I will prioritize students who are there with questions, so help expedite this process I ask that you let me know in advance which Learning Target you want to re-assess. You must also bring a re-assessment ticket with you (see below).
- You may schedule a re-assessment using my Calendly link. If there's a time you're looking for that you don't see available on Calendly, just let me know!
- Re-assessments done outside of class may be completed either in written form or through conversation with me.

Finally, please note the following restrictions on re-assessing:

- You may not attempt more than two (2) re-assessments outside of class per week.
- You may not attempt to re-assess more than one learning target per day. Exceptions to this may be made in certain cases where Learning Targets are closely aligned.
- No re-assessments, other than the final check-in, will be given during finals week.

0.7.3 Re-assessment Ticket

If you are reassessing a learning target outside of class, you must submit written responses, on a separate piece of paper, to the following questions at the time of your reassessment:

1. What were your mistakes/gaps/sticking points on your previous attempt?
2. What steps did you take to improve your understanding? Did you re-do the problem? List any resources you used to study and prepare for this attempt.

0.8 Midterm and Final Check-Ins

Instead of a traditional midterm or final exam, we will instead have a longer check-in which covers all of the Learning Targets that we've covered up to that point. These are not worth anything more –

they are simply **additional** opportunities to demonstrate proficiencies. If you've already done that to your satisfaction, then these are **optional**!

Midterm Check-In

- Tuesday, October 6th

Final Check-In

- Monday, Dec. 7th, 1:30-3:30

0.9 Grades

0.10 Course Level Learning Objectives

By the end of the semester you will be able to...

- Analyze functions defined graphically, numerically, or by formula, including exponential, logarithmic, trigonometric, polynomial, and rational families of functions.
- Perform algebraic computations efficiently and correctly; particularly on exponential, logarithmic, and fractional expressions.
- Construct graphs of trigonometric functions using the properties of the unit circle.
- Use trigonometric, rational, exponential, and logarithmic equations to model situations and solve problems.
- Develop persistence in the face of challenging problems and reflective practices to improve your learning.
- Communication mathematics effectively using appropriate justifications and representations.

1 Reminders and Expectations

- Canvas Consider Canvas your "base camp" for this course. All information, assignments, updates, Zoom links, etc. will be posted there. If you have a question, it's probably answered on Canvas (perhaps in this very syllabus...). I expect you to check the site frequently to stay on top of deadlines and course communications.
- I believe in you. Doubting yourself? It's human. Need an affirmation? Let me know... I've got your back!